College of Computing
CS & AI Department
Advanced Deep
Learning Course No:
AI4203

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1. Dataset Source and Definition

Dataset Name: Rotten Tomatoes Movie Reviews [1].
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Dataset Definition: This dataset contains 10,662 balanced Rotten Tomatoes movie reviews (5,331 positive and 5,331 negative), where each entry includes a text field and a corresponding sentiment label.

2. Task Definition and Target Label

The task is binary sentiment classification where each movie review is mapped to a discrete label (1 for positive, 0 for negative) based on its semantic.

3. Input Representation Plan

Create a custom pipeline mapping unique words from training to a discrete integer IDs. Using the "PAD" and "UNK" tokens to handle variable sequence length and out-of-vocabulary tokens in a torch.LongTenso inputs.

4. Split Plan and Leakage Prevention

Splitting the data by 80% training (8530 samples), 10% validation (1066 samples), and 10% testing (1066 samples), to make sure the vocabulary is derived exclusively from the training set to prevent data leakage, by building the vocabulary only from training set and use "UNK" for new words.

5. Primary Metric with Justification (Accuracy vs Macro-F1, etc.)

Since the data we are working with is balanced, we will be using the accuracy metric to measure the classification performance. it works well with binary sentiments and provides a clear measure of how often the model identifies them correctly [4].

6. Compute and Time Budget Constraints

The computational cost is dominated by the self-attention mechanism, which scales as $O\ N^2 \cdot D$.
To remain feasible on a CPU-only Google Colab setup, the sequence length is capped at $N_X = 50$, and a dry run with a single encoder block, a small batch size (≤ 16), and 5 epochs completes within a few seconds.

7. Anticipated Risk and Mitigation Strategy

Risk: Noise and Overfitting in Short-Text Sequences

Short movie reviews may contain noisy or sentiment-heavy tokens, increasing the risk of overfitting. Dropout with a rate of 0.1 is applied in both the self-attention and feed-forward layers to improve generalization [2]. However,some degree of overfitting may persist due to the short sequence length, limited data, and the high representational capacity of Transformer-based models. [3].

8. Full training and evaluation

Table 1: Baseline Candidate Results for Transformer Sentiment Classifier

Run #	Batch Size	d_{model}	Learning Rate	Best Epoch	Val Loss (best)	Val Acc	Test Acc	Test Loss
1	32	64	1×10^{-4}	6	0.6629	0.608	0.597	0.668
2	32	64	5×10^{-4}	4	0.6455	0.647	0.649	0.637
3	32	128	1×10^{-4}	4	0.6461	0.611	0.631	0.637
4	**32**	**128**	$\mathbf{5 \times 10^{-4}}$	**6**	**0.6153**	**0.669**	**0.705**	**0.575**
5	64	64	1×10^{-4}	14	0.6690	0.601	0.624	0.644
6	64	64	5×10^{-4}	5	0.6571	0.623	0.654	0.623
7	64	128	1×10^{-4}	9	0.6470	0.627	0.637	0.623
8	64	128	5×10^{-4}	3	0.6469	0.618	0.651	0.608

The model was successfully trained on the complete dataset for 15 epochs using the Transformer-based classifier. We have tried 8 training candidates, and the best training used the following hyperparameters: batch size = 32, learning rate = 5×10^{-4}, model dimension = 128, 8 attention heads, and 1 encoder block.

The training ran smoothly, and early stopping activated at epoch 8 when the validation loss stopped improving (patience = 2). The final recorded metrics were:

- Best Validation Accuracy: 0.669
- Best Validation Loss: 0.615
- Test Accuracy: 0.705
- Test Loss: 0.575

The loss decreased steadily across epochs, while accuracy improved for both training and validation sets. The confusion matrix shows balanced performance between positive and negative classes, with no major label bias observed.

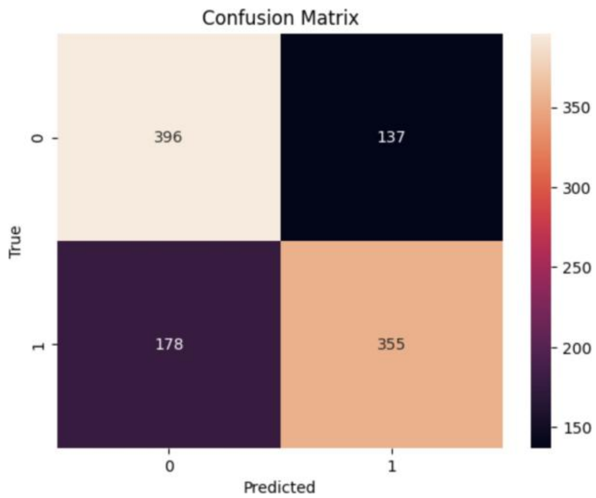

Figure 1: Baseline confusion matrix.

Overall, the model trains correctly and generalizes reasonably well to the held-out test set, confirming that the Transformer encoder and training loop were implemented properly providing a solid baseline comparable to classical approaches (NLTK Naive Bayes: 77.38%, Scikit-learn SVM: 78%), consistent with established benchmarks for sentiment analysis on same datasets [7].

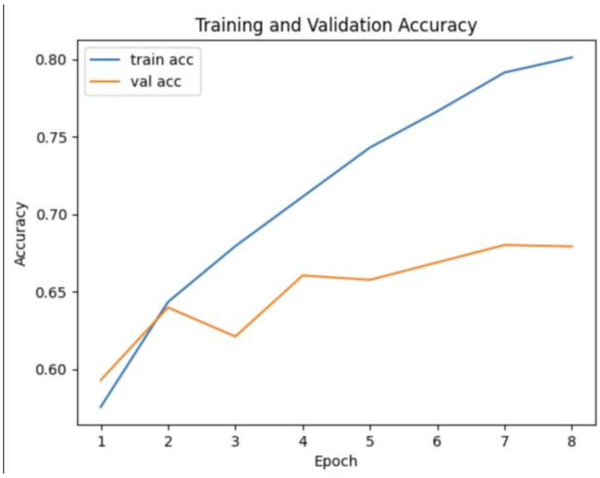

Figure 2: Baseline accuracy results.

Figure 3: Baseline loss curve.

9. Ablation study

The ablation study 1: investigates the effect of varying attention heads (see Table 2).

Table 2: Ablation Study Results: Number of Attention Heads

Model	Best Epoch	Train Loss	Train Acc	Val Loss	Val Acc	Test Loss	Test Acc	TN	FP	FN	TP
H=8 (Baseline)	6	0.4856	0.766	0.6153	0.669	0.575	0.705	396	137	178	355
H=4	7	0.4566	0.779	0.6109	0.679	0.5861	0.7214	374	159	138	395
H=2	7	0.4706	0.774	0.5932	0.685	0.5680	0.7092	350	183	127	406

Using fewer heads for this task improved the Val accuracy, as they suit the data simplicity. However, the performance gaps compared to the baseline and between the heads themselves were very small. Given these marginal differences, two experiments are not enough to reach a definitive conclusion.

The ablation study 2: investigates the effect of varying dropout values (see Table 3).

Table 3: Ablation Study Results: Dropout Rate

Model	Best Epoch	Train Loss	Train Acc	Val Loss	Val Acc	Test Loss	Test Acc	True Neg	False Pos	False Neg	True Pos
D=0.1 (Baseline)	6	0.4856	0.766	0.6153	0.669	0.575	0.705	396	137	178	355
D=0.3	4	0.6326	0.640	0.6568	0.611	0.6462	0.6266	223	310	88	445

Increasing dropout to (0.3) introduced excessive regularization, leading to underfitting and lower overall performance. This suggests that the higher dropout rate is too strong for the dataset, while a moderate rate (0.1) provides better generalization.

10. Error analysis

Sarcasm and nuanced critiques represent two primary categories of sentiment analysis misclassifications. These errors typically manifest in the following ways:

- **Semantic Inversion Failure:** Misclassifications occur when reviewers employ negative terminology to describe positive attributes (e.g., *"terrifyingly good"*, *"wickedly funny"*). A shallow self-attention mechanism often fails to invert the polarity of these tokens correctly. [5].

- **Ironic Connotations:** Errors arise when positive vocabulary is used with ironic intent to mask an ultimately negative conclusion. [5].

Conclusion: This suggests that the current architecture's self-attention structure lacks the sufficient depth required to evaluate the global contextual sentiment against localized, misleading tokens.

Table 4: Analysis of Misclassified Text Samples (False Negatives)

Index	Text Content	True	Pred
0	lovingly photographed in the manner of a golden book sprung to life , stuart little 2 manages sweetness largely without stickiness .	1	0
2	it's like a big chill reunion of the baader meinhof gang , only these guys are more harmless pranksters than political activists .	1	0
3	the story gives ample opportunity for large scale action and suspense , which director shekhar kapur supplies with tremendous skill .	1	0
4	red dragon never cuts corners .	1	0
6	throws in enough clever and unexpected twists to make the formula feel fresh .	1	0
7	weighty and ponderous but every bit as filling as the treat of the title .	1	0
9	generates an enormous feeling of empathy for its characters .	1	0
10	exposing the ways we fool ourselves is one hour photo's real strength .	1	0

11. Responsible AI mini-note

This project utilizes the **Rotten Tomatoes Movie Reviews** dataset to train an automated sentiment analysis system. During the evaluation of the model's performance, the following risks and mitigation strategies were identified:

1. **Representational Bias:** The dataset heavily represents professional critics. Their structured vocabulary may cause the model to misinterpret informal dialects, slang, or cultural idioms used by general audiences.
2. **Data Augmentation:** To mitigate this bias, the training corpus should be augmented with diverse, user-generated reviews to enhance linguistic fairness and model generalizability.
3. **Confidence Thresholding:** Establishing a confidence threshold can flag highly ambigu-ous or sarcastic text for human review, preventing the enforcement of incorrect automated classifications. [6].

References

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